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Roll No .....

**BE-3001 (CS/IT)-CBGS**

**B.E., III Semester**

Examination, June 2020

**Choice Based Grading System (CBGS)**

**Mathematics - III**

**Time : Three Hours**

**Maximum Marks : 70**

- Note:** i) Attempt any five questions.  
ii) All questions carry equal marks.

1. a) Find the Fourier series for  $f(x) = x^2$  in  $-\pi < x < \pi$ .  
b) Express  $f(x) = x$  as a half range sine series in  $0 < x < 2$ .
2. a) Find the Fourier transform of  $f(x) = \begin{cases} x, & |x| \leq a \\ 0, & |x| > a \end{cases}$   
b) Find the Fourier cosine transform of the  
$$f(x) = e^{-3x} + e^{-4x}.$$
3. a) Find  $L\{\sin 3t \cos 2t + t^{3/2} - 5e^{-2t} + 6 \cos 3t\}$   
b) Find  $L\{e^{-4t}(\sin 2t + \cos 2t)\}$

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4. a) Find  $L\left\{\frac{1 - \cos 2t}{t}\right\}$

b) Evaluate  $\int_0^{\infty} t e^{-3t} \sin t \, dt$

5. a) Evaluate  $L^{-1}\left\{\frac{6s^2 + 22s + 18}{(s+1)(s+2)(s+3)}\right\}$

b) Solve  $\frac{d^2y}{dt^2} + 6\frac{dy}{dt} + 9y = \sin t$ , given that  $y=1, \frac{dy}{dt} = 0$  when  $t = 0$ .

6. a) If  $f(x) = cx^2$ ,  $0 < x < 1$ , find the value of C and determine the probability that  $\frac{1}{3} < x < \frac{1}{2}$ .

b) A continuous random variable X has the density function  $f(x) = 3x^2$ ,  $0 \leq x < 1$ , find a and b when

i)  $P(X \leq a) = P(X > a)$

ii)  $P(X > b) = 0.05$ .

7. a) Six dice are thrown 729 times. How many times do you expect at least three dice to show a five or six?

b) Fit Poisson's distribution to the following and calculate theoretical frequencies ( $e^{-0.5} = 0.61$ ):

|            |     |    |    |   |   |
|------------|-----|----|----|---|---|
| Deaths:    | 0   | 1  | 2  | 3 | 4 |
| Frequency: | 122 | 60 | 15 | 2 | 1 |

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8. a) Find the mean and variance of the Binomial distribution.  
b) Fit a second degree parabola to the following data regarding  $x$  as an independent variable:

|       |   |   |    |    |    |
|-------|---|---|----|----|----|
| $x :$ | 0 | 1 | 2  | 3  | 4  |
| $y :$ | 1 | 5 | 10 | 22 | 38 |

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